

Big Data: Lab 1

General

I used an Apache Spark Docker image to perform the work. MacOS was used.

Main steps

I pulled the Spark image from Docker Hub. I created and ran the Spark Master container. I created and ran one Spark Worker container. I linked the worker to the master. I verified the cluster and the session run using the Spark Web UI.

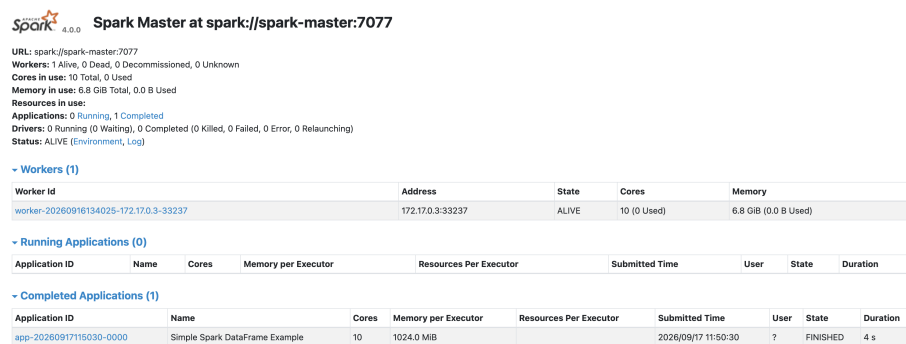


Figure 1: Apache Spark WebUI showing the worker and the completed application.

Encountered errors

First error

The cause: bitnami's spark image is no longer being updated and have therefore been moved to bitnamilegacy/spark on docker hub.

```
(spark) wilmer@MacBook-Pro-som-tillhor-Wilmer Exercise Files % docker pull bitnami/spark
Using default tag: latest
Error response from daemon: failed to resolve reference "docker.io/bitnami/spark:latest": docker.io/bitnami/spark:lates
t: not found
```

Figure 2: Image of first error in terminal.

The solution: Change the name from bitnami/spark to bitnamilegacy/spark in the command: `docker pull bitnamilegacy/spark`.

Also adding "legacy" in the following commands.

```
docker run -d --name spark-master -h spark-master -p 8080:8080 bitnamilegacy/spark
```

```
docker run -d --name spark-worker -h spark-worker --link spark-master:spark-master bitnamilegacy/spark
```

Second error

While attempting to run the spark-submit command:

```
/opt/bitnami/spark/bin/spark-submit --master spark://spark-master:7077 /opt/bitnami/spark/s
```

I got the following error:



```
I have no name!@spark-master:/opt/bitnami/spark$ /opt/bitnami/spark/bin/spark-submit --master spark://spark-master:7077 /opt/bitnami/spark/simple_dataframe.py
WARNING: Using incubator modules: jdk.incubator.vector
Exception in thread "main" java.lang.IllegalArgumentException: basedir must be absolute: ../ivy2.5.2/local
    at org.apache.ivy.util.Checks.checkAbsolute(Checks.java:48)
    at org.apache.ivy.plugins.repository.file.FileRepository.setBaseDir(FileRepository.java:137)
    at org.apache.ivy.plugins.repository.file.FileRepository.<init>(FileRepository.java:44)
    at org.apache.spark.util.MavenUtils$.createRepoResolvers(MavenUtils.scala:159)
    at org.apache.spark.util.MavenUtils$.buildIvySettings(MavenUtils.scala:287)
    at org.apache.spark.util.DependencyUtils$.resolveMavenDependencies(DependencyUtils.scala:118)
    at org.apache.spark.deploy.SparkSubmit.prepareSubmitEnvironment(SparkSubmit.scala:341)
    at org.apache.spark.deploy.SparkSubmit.org$apache$spark$deploy$SparkSubmit$$runMain(SparkSubmit.scala:961)
    at org.apache.spark.deploy.SparkSubmit.doRunMain$1(SparkSubmit.scala:284)
    at org.apache.spark.deploy.SparkSubmit.submit(SparkSubmit.scala:227)
    at org.apache.spark.deploy.SparkSubmit.doSubmit(SparkSubmit.scala:96)
    at org.apache.spark.deploy.SparkSubmit$$anon$2.doSubmit(SparkSubmit.scala:1132)
    at org.apache.spark.deploy.SparkSubmit$.main(SparkSubmit.scala:1141)
    at org.apache.spark.deploy.SparkSubmit.main(SparkSubmit.scala)
```

Figure 3: Image of second error in terminal.

solution: Add the flag `-conf` to give an actual address for the `.ivy` directory:

```
/opt/bitnami/spark/bin/spark-submit \
--master spark://spark-master:7077 \
--conf spark.jars.ivy=/tmp/.ivy2 \
/opt/bitnami/spark/simple_dataframe.py
```

Third error

Running the updated command gave me a new error:

```
WARNING: Using incubator modules: jdk.incubator.vector
File "/opt/bitnami/spark/simple_dataframe.py", line 6
    .appName("Simple Spark DataFrame Example")
IndentationError: unexpected indent
26/09/17 11:41:18 INFO ShutdownHookManager: Shutdown hook called
26/09/17 11:41:18 INFO ShutdownHookManager: Deleting directory /tmp/spark-6160c04f-bc29-4a15
```

This was solved by adjusting line 6 in the `simple_dataframe.py` file to have the three consecutive functions to be on the same line:

```
spark = SparkSession.builder.appName("Simple Spark DataFrame Example").getOrCreate()
```